

Fundamentals of RF and Microwave Noise Figure Measurements

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What is Noise Figure?

Modern receiving systems must often process very weak signals, but the noise added by the system components tends to obscure those very weak signals. Sensitivity, bit error ratio (BER) and noise figure are system parameters that characterize the ability to process low-level signals. Of these parameters, noise figure is unique in that it is suitable not only for characterizing the entire system but also the system components such as the pre-amplifier, mixer, and IF amplifier that make up the system. By controlling the noise figure and gain of system components, the designer directly controls the noise figure of the overall system. Once the noise figure is known, system sensitivity can be easily estimated from system bandwidth. Noise figure is often the key parameter that differentiates one system from another, one amplifier from another, and one transistor from another. Such widespread application of noise figure specifications implies that highly repeatable and accurate measurements between suppliers and their customers are very important.

The other approach is to minimize the noise generated within receiver components. Noise measurements are key to assuring that the added noise is minimal. Once noise joins the signals, receiver components can no longer distinguish noise in the signal frequency band from legitimate signal fluctuations. The signal and noise get processed together. Subsequent raising of the signal level with gain, for example, will raise the noise level an equal amount.



Keysight analyzers simplify noise figure measurements

This application note is part of a series about noise measurement. Much of what is discussed is either material that is common to most noise figure measurements or background material. It should prove useful as a primer on noise figure measurements. The need for highly repeatable, accurate and meaningful measurements of noise without the complexity of manual measurements and calculations has led to the development of noise figure measurement instruments with simple user interfaces. Using these instruments does not require an extensive background in noise theory. A little noise background may prove helpful, however, in building confidence and understanding a more complete picture of noise in RF and microwave systems. Other literature to consider for additional information on noise figure measurements is indicated throughout this note. Numbers appearing throughout this document in square brackets [] correspond to the same numerical listing in the References section. Related Keysight Technologies literature and web resources appear later in this application note.

The reason for measuring noise properties of networks is to minimize the problem of noise generated in receiving systems. One approach to overcome noise is to make the weak signal stronger. This can be accomplished by raising the signal power transmitted in the direction of the receiver, or by increasing the amount of power the receiving antenna intercepts, for example, by increasing the aperture of the receiving antenna. Raising antenna gain, which usually means a larger antenna, and raising the transmitter power, are eventually limited by government regulations, engineering considerations, or economics.

The Importance of Noise in Communication Systems

The signal-to-noise (S/N) ratio at the output of receiving systems is a very important criterion in communication systems. Detecting signals in the presence of noise is a fundamental challenge in communication systems. While early systems often relied on human interpretation of audio signals, modern systems quantify performance using objective metrics such as signal-to-noise ratio, bit error ratio (BER), and sensitivity. Noise figure and sensitivity are measurable and objective figures of merit. Noise figure and sensitivity are closely related (see Sensitivity in the glossary). For digital communication systems, a quantitative reliability measure is often stated in terms of BER or the probability that any received bit is in error. BER is related to noise figure in a non-linear way. As the S/N ratio decreases gradually, for example, the BER increases suddenly near the noise level where 1s and 0s become confused. Noise figure characterizes the inherent noise performance of the system, while BER reflects the resulting communication reliability.

Figure 1-1, which shows the probability of error versus carrier- to-noise ratio for several types of digital modulation, indicates that BER changes by several orders of magnitude for only a few dB change in signal-to-noise ratio.

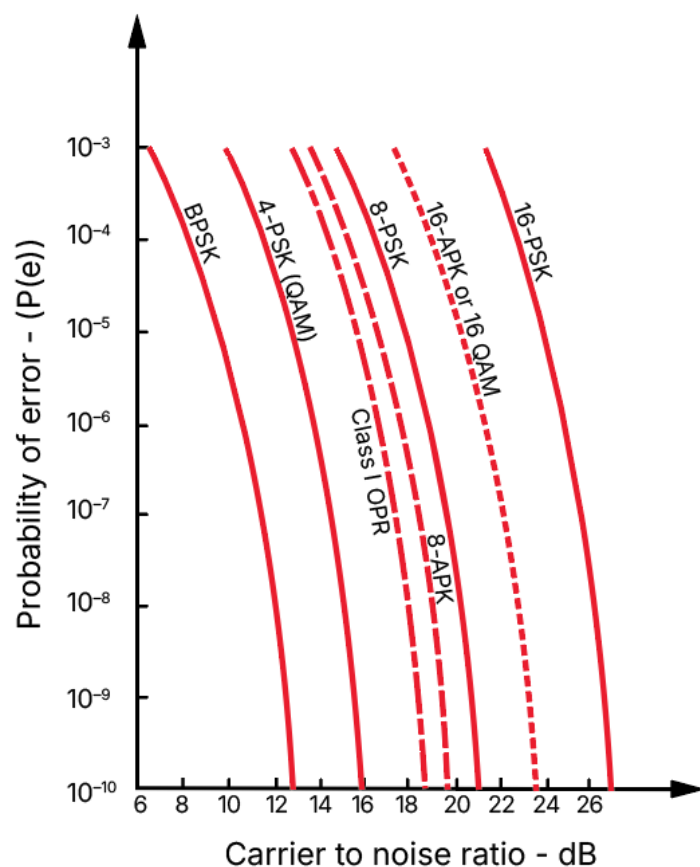


Figure 1-1. Probability of error, $P(e)$, as a function of carrier-to-noise ratio (C/N, which can be interpreted as signal-to-noise ratio), for various kinds of digital modulation. From Kamilo Feher, DIGITAL COMMUNICATIONS: Microwave Applications, ©1981, p.71. Reprinted by permission of Prentice-Hall, Inc., Englewood Cliffs, NJ

In satellite systems, noise figure may be particularly important. Consider the example of lowering the noise figure of a receiver in a satellite communication system, such as a direct broadcast satellite (DBS) receiver. One option for improving receiver noise figure is to increase the transmitter power, however, this option can be very costly to implement. A better alternative is to substantially improve the performance of the receiver low noise amplifier (LNA). It is easier to improve LNA performance than to increase transmitter power.



DBS receiver

The reason for measuring noise properties of networks is to minimize the problem of noise generated in receiving systems. One approach to overcome noise is to make the weak signal stronger. This can be accomplished by raising the signal power transmitted in the direction of the receiver, or by increasing the amount of power the receiving antenna intercepts, for example, by increasing the aperture of the receiving antenna. Raising antenna gain, which usually means a larger antenna, and raising the transmitter power, are eventually limited by government regulations, engineering considerations, or economics.

In the case of a production line that produces satellite receivers, it may be quite easy to reduce the noise figure 1 dB by adjusting impedance levels or carefully selecting specific transistors. A 1 dB reduction in noise figure has approximately the same effect as increasing the antenna diameter by 40 percent. But increasing the diameter could change the design and significantly raise the cost of the antenna and support structure.

Sometimes noise is an important parameter of transmitter design. For example, if a linear, broadband, power amplifier is used on a base station, excess broadband noise could degrade the signal-to-noise ratio at the adjacent channels and limit the effectiveness of the system. The noise figure of the power amplifier could be measured to provide a figure of merit to ensure acceptable noise levels before it is installed in the system.

Sources of Noise

The noise being characterized by noise measurements consists of spontaneous fluctuations caused by ordinary phenomena in the electrical equipment. Thermal noise arises from vibrations of conduction electrons and holes due to their finite temperature. Some of the vibrations have spectral content within the frequency band of interest and contribute noise to the signals. The noise spectrum produced by thermal noise is nearly uniform over RF and microwave frequencies. The power delivered by a thermal source into an impedance matched load is kTB watts, where k is Boltzmann's constant (1.38×10^{-23} joules/K), T is the temperature in K, and B is the system's noise bandwidth. The available power is independent of the source impedance. The available power into a matched load is directly proportional to the bandwidth so that twice the bandwidth would allow twice the power to be delivered to the load. (See thermal noise in the glossary.)

Shot noise arises from the quantized nature of current flow (see shot noise in the glossary). Other random phenomena occur in nature that are quantized and produce noise in the manner of shot noise. Examples are the generation and recombination of hole/electron pairs in semiconductors (G-R noise), and the division of emitter current between the base and collector in transistors (partition noise). These noise generating mechanisms have the characteristic that like thermal noise, the frequency spectrum is essentially uniform, producing equal power density across the entire RF and microwave frequency range.

There are many causes of random noise in electrical devices. Noise characterization usually refers to the combined effect from all the causes in a component. The combined effect is often referred to as if it all were caused by thermal noise. Referring to a device as having a certain noise temperature does not mean that the component is that physical temperature, but merely that its noise power is equivalent to a thermal source of that temperature. Although the noise temperature does not directly correspond with physical temperature there may be a dependence on temperature. Some very low noise figures can be achieved when the device is cooled to a temperature below ambient.

Noise as referred to in this application note does not include human-generated interference, although such interference is very important when receiving weak signals. This note is not concerned with noise from ignition, sparks, or with undesired pick-up of spurious signals. Nor is this note concerned with erratic disturbances like electrical storms in the atmosphere. Such noise problems are usually resolved by techniques like relocation, filtering, and proper shielding. Yet these sources of noise are important here in one sense—they upset the measurements of the spontaneous noise this note is concerned with. A manufacturer of LNAs may have difficulty measuring noise figure due to unintended RF interference from nearby transmitters, such as base stations, or other signals operating within the measurement frequency range. For this reason, accurate noise figure measurements are often performed in shielded rooms.

The Concept of Noise Figure

The most basic definition of noise figure came into popular use in the 1940's when Harold Friis^[8] defined the noise figure F of a network to be the ratio of the signal-to-noise power ratio at the input to the signal-to-noise power ratio at the output.

$$F = \frac{S_i/N_i}{S_o/N_o} \quad (1-1)$$

Thus, the noise figure of a network is the decrease or degradation in the signal-to-noise ratio as the signal goes through the network. A perfect amplifier would amplify the noise at its input along with the signal, maintaining the same signal-to-noise ratio at its input and output (the source of input noise is often thermal noise associated with the earth's surface temperature or with losses in the system). A realistic amplifier, however, also adds some extra noise from its own components and degrades the signal-to-noise ratio. A low noise figure means that very little noise is added by the network. The concept of noise figure only fits networks (with at least one input and one output port) that process signals. This note is mainly about two-port networks; although mixers are in general three-port devices, they are usually treated the same as a two-port device with the local oscillator connected to the third port.

It might be worthwhile to mention what noise figure does not characterize. Noise figure is not a quality factor of networks with one port; it is not a quality factor of terminations or of oscillators. Oscillators have their own quality factors like carrier-to-noise ratio and phase noise. But receiver noise generated in the sidebands of the local oscillator driving the mixer, can get added by the mixer. Such added noise increases the noise figure of the receiver.

Noise figure has nothing to do with modulation or demodulation. It is independent of the modulation format and of the fidelity of modulators and demodulators. Noise figure is, therefore, a more general concept than noise-quieting used to indicate the sensitivity of FM receivers or BER used in digital communications.

Noise figure should be thought of as separate from gain. Once noise is added to the signal, subsequent gain amplifies signal and noise together and does not change the signal-to-noise ratio.

Figure 1-2(a) shows an example situation at the input of an amplifier. The depicted signal is 40 dB above the noise floor: Figure 1-2(b) shows the situation at the amplifier output. The amplifier's gain has boosted the signal by 20 dB. It also boosted the input noise level by 20 dB and then added its own noise. The output signal is now only 30 dB above the noise floor. Since the degradation in signal-to-noise ratio is 10 dB, the amplifier has a 10 dB noise figure.

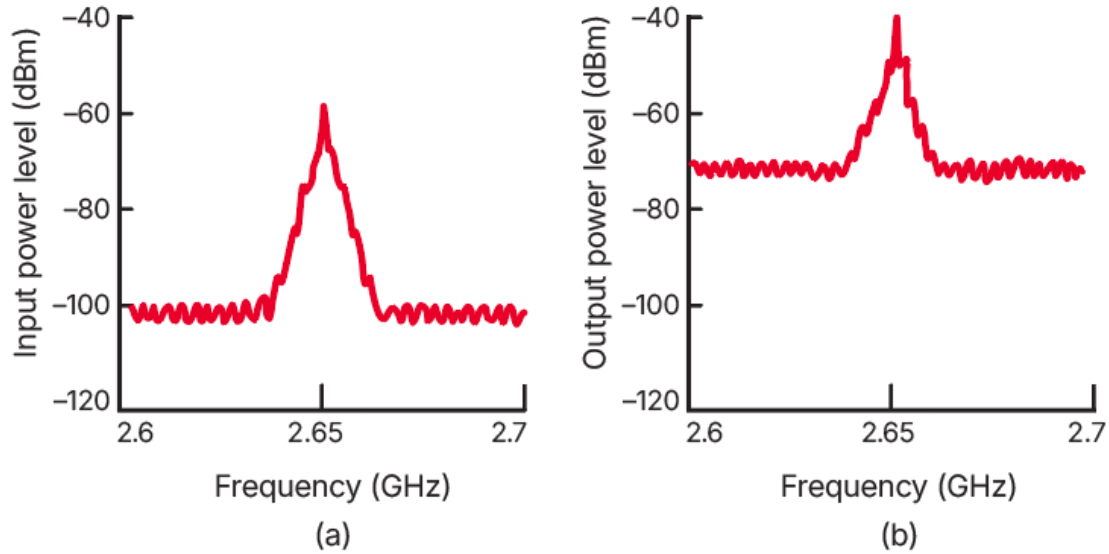


Figure 1-2. Typical signal and noise levels versus frequency (a) at an amplifier's input and (b) at its output. Note that the noise level rises more than the signal level due to added noise from amplifier circuits. This relative rise in noise level is expressed by the amplifier noise figure.

Note that if the input signal level were 5 dB lower (35 dB above the noise floor) it would also be 5 dB lower at the output (25 dB above the noise floor), and the noise figure would still be 10 dB. Thus, noise figure is independent of the input signal level.

A more subtle effect will now be described. The degradation in a network's signal-to-noise ratio is dependent on the temperature of the source that excites the network. This can be proven with a calculation of the noise figure F , where S_i and N_i represent the signal and noise levels available at the input to the device under test (DUT), S_o and N_o represent the signal and noise levels available at the output, N_a is the noise added by the DUT, and G is the gain of the DUT. Equation (1-2) shows the dependence on noise at the input N_i .

$$\begin{aligned}
 F &= \frac{S_i/N_i}{S_o/N_o} \\
 &= \frac{S_i/N_i}{GS_i/(N_a + GN_i)} \\
 &= \frac{N_a + GN_i}{GN_i}
 \end{aligned} \tag{1-2}$$

The input noise level is usually thermal noise from the source and is referred to by kT_0B . Friis^[8] suggested a reference source temperature of 290 K (denoted by T_0), which is equivalent to 16.8°C and 62.3°F. This temperature is close to the average temperature seen by receiving antennas directed across the atmosphere at the transmitting antenna.

The power spectral density kT_0 , furthermore, is the even number 4.00×10^{-21} watts per hertz of bandwidth (-174 dBm/Hz). The IRE (forerunner of the IEEE) adopted 290 K as the standard temperature for determining noise figure^[7]. Then equation (1-2) becomes:

$$F = \frac{N_a + kT_0 BG}{kT_0 BG} \quad (1-3)$$

This is the definition of noise figure adopted by the IRE.

Noise figure is generally a function of frequency, but it is usually independent of bandwidth (so long as the measurement bandwidth is narrow enough to resolve variations with frequency). Noise powers N_a and N_i of equation (1-2) are each proportional to bandwidth. But the bandwidth in the numerator of (1-2) cancels with that of the denominator—resulting in noise figure being independent of bandwidth.

In summary, the noise figure of a DUT is the degradation in the signal-to-noise ratio as a signal passes through the DUT. The specific input noise level for determining the degradation is that associated with a 290 K source temperature. The noise figure of a DUT is independent of the signal level so long as the DUT is linear (output power versus input power).

The IEEE Standard definition of noise figure, equation (1-3), states that noise figure is the ratio of the total noise power output to that portion of the noise power output due to noise at the input when the input source temperature is 290 K.

While the quantity F in equation (1-3) is often called noise figure, more often it is called noise factor or sometimes noise figure in linear terms. Modern usage of noise figure usually is reserved for the quantity NF , expressed in dB units:

$$NF = 10 \log F \quad (1-3)$$

This is the convention used in the remainder of this application note.

Noise Figure and Noise Temperature

Sometimes effective input noise temperature (T_e) is used to describe the noise performance of a device rather than the noise figure (NF). Quite often temperature units are used for devices used in satellite receivers. T_e is the equivalent temperature of a source impedance into a perfect (noise-free) device that would produce the same added noise, N_a . It is often defined as

$$T_e = \frac{N_a}{kB} \quad (1-4)$$

It can be related to the noise factor F:

$$T_e = T_0(F-1), \text{ where } T_0 \text{ is } 290 \text{ K} \quad (1-5)$$

The input noise level present in terrestrial VHF and microwave communications is often close to the 290 K reference temperature used in noise figure calculations due to the earth's surface temperature. When this is the case, a 3 dB change in noise figure will result in a 3 dB change in the signal-to-noise ratio.

In satellite receivers the noise level coming from the antenna can be far less, limited by sidelobe radiation and the background sky temperature to values often below 100 K. In these situations, a 3 dB change in the receiver noise figure may result in much more than 3 dB signal-to-noise change. While system performance may be calculated using noise figure without any errors (the 290 K reference temperature need not correspond to actual temperature), system designers may prefer to use T_e as a system parameter.

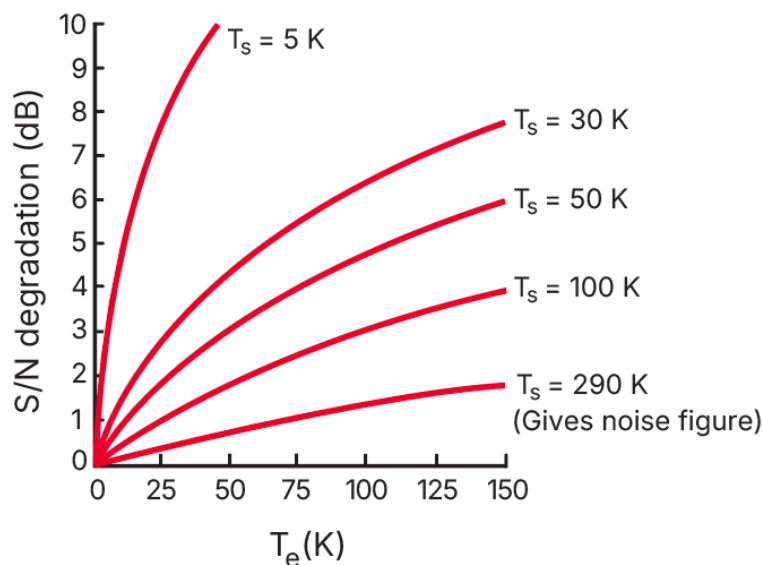


Figure 1-3. Degradation in the S/N ratio versus T_e of a device for various values of temperature for the source impedance. Noise figure is defined for a source temperature of 290 K.

Noise Characteristics of Two-Port Networks

The Noise Figure of Multi-Stage Systems

The noise figure definition covered in Chapter 1 can be applied to both individual components such as a single transistor amplifier, or to a complete system such as a receiver. The overall noise figure of the system can be calculated if the individual noise figures and gains of the system components are known. To find the noise figure of each component in a system, the internal noise added by each stage, N_a , must be found. The gain must also be known. The actual methods used to determine noise and gain are covered in Chapter 3, The Measurement of Noise Figure. The basic relationship between the individual components and the system will be discussed here.

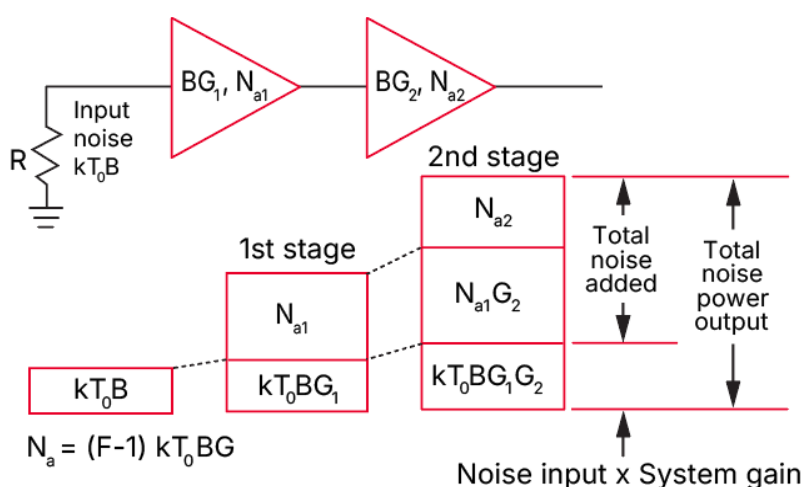


Figure 2-1. The effect of second stage contribution.

For two stages, see Figure 2-1, the output noise will consist of the kT_0B source noise amplified by both gains, G_1G_2 , plus the first amplifier output noise, N_{a1} , amplified by the second gain, G_2 , plus the second amplifiers output noise, N_{a2} . The noise power contributions may be added since they are uncorrelated. Using equation (1-3) to express the individual amplifier noise contributions, the output noise can be expressed in terms of their noise factors, F .

$$N_o = kT_0BG_1G_2 \left[F_1 + \frac{F_2 - 1}{G_1} \right] \quad (2-1)$$

With the output noise known, the noise factor of the combination of both amplifiers can be calculated using equation (1-1). This is the overall system noise figure of this two-stage example. The quantity $(F_2-1)/G_1$ is often called the second stage contribution. One can see that as long as the first stage gain is high, the second stage contribution will be small. This is why the pre-amplifier gain is an important parameter in receiver design.

$$F_{\text{sys}} = F_1 + \frac{F_2 - 1}{G_1} \quad (2-2)$$

Equation (2-2) can be re-written to find F_1 if the gain and overall system noise factor are known. This is the basis of corrected noise measurements and will be discussed in the next chapter.

This calculation may be extended to an n-stage cascade of devices and expressed as Equation (2-3) is often called the cascade noise equation.

$$F_{\text{sys}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots + \frac{F_n - 1}{G_1 G_2 \dots G_{n-1}} \quad (2-3)$$

Gain and Mismatch

The device gain is an important parameter in noise calculations. When an input power of kT_0B is used in these calculations, it is an available power, the maximum that can be delivered to a matched load. If the device has a large input mismatch (not unusual for low-noise amplifiers), the actual power delivered to the device would be less. If the gain of the device is defined as the ratio of the actual power delivered to the load to the maximum power available from the source, we can ignore the mismatch loss present at the input of the device since it is taken into account in our gain definition. This definition of gain is called transducer gain, G_t . When cascading devices, however, mismatch errors arise if the input impedance of the device differs from the load impedance. In this case the total gain of a cascaded series of devices does not equal the product of the gains.

Available gain, (G_a), is often given as a transistor parameter, it is the gain that will result when a given source admittance, Y_s , drives the device and the output is matched to the load. It is often used when designing amplifiers. Refer to the glossary for a more complete description of the different definitions of gain.

Most often insertion gain, G_i , or the forward transmission coefficient, $(S_{21})^2$, is the quantity specified or measured for gain in a 50Ω system. If the measurement system has low reflection coefficients and the device has a good output match there will be little error in applying the cascade noise figure equation (2-3) to actual systems. If the device has a poor output match or the measurement system has significant mismatch errors, an error between the actual system and calculated performance will occur. If, for example, the output impedance of the first stage was different from the 50Ω source impedance that was used when the second stage was characterized for noise figure, the noise generated in the second stage could be altered. Fortunately, the second stage noise contribution is reduced by the first stage gain so that in many applications errors involving the second stage are minimal. When the first stage has low gain (G^2F_2), second stage errors can become significant. The complete analysis of mismatch effects in noise calculations is lengthy and generally requires understanding the dependence of noise figure on source impedance. This effect, in addition to the gain mismatch effect, will be discussed in the next section (Noise Parameters). It is because of this noise figure dependence that S-parameter correction is not as useful as it would seem in removing the errors associated with mismatch.^[4]

Noise Parameters

Noise figure is, in principle, a simplified model of the actual noise in a system. A single, theoretical noise element is present in each stage. Most actual amplifying devices such as transistors can have multiple noise contributors; thermal, shot, and partition as examples. The effect of source impedance on these noise generation processes can be a very complex relationship. The noise figure that results from a noise figure measurement is influenced by the match of the noise source and the match of the measuring instrument; the noise source is the source impedance for the DUT, and the DUT is the source impedance for the measuring instrument. The actual noise figure performance of the device when it is in its operating environment will be determined by the match of other system components.

Designing low noise amplifiers requires tradeoffs between the gain of a stage and its corresponding noise figure. These decisions require knowledge of how the active device's gain and noise figure change as a function of the source impedance or admittance. The minimum noise figure does not necessarily occur at either the system impedance, Z_o , or at the conjugate match impedance that maximizes gain.

To fully understand the effect of mismatch in a system, two characterizations of the device-under-test (DUT) are needed, one for noise figure and another for gain. While S-parameter correction can be used to calculate the available gain in a perfectly matched system, it cannot be used to find the optimum noise figure. A noise parameter characterization uses a special tuner to present different complex impedances to the DUT.^[29]

The dependence of noise factor on source impedance presented by the tuner is described by

$$F = F_{\min} + \frac{4R_n}{Z_o} \left(\frac{|\Gamma_{\text{opt}} - \Gamma_s|^2}{|1 + \Gamma_{\text{opt}}|^2 (1 - |\Gamma_s|^2)} \right) \quad (2-4)$$

where the Γ is the source reflection coefficient that results in the noise factor F . In the equation, $F_{\min}$ is the minimum noise factor for the device that occurs when $\Gamma = \Gamma_{\text{opt}}$. R_n is the noise resistance (the sensitivity of noise figure to source admittance changes). $F_{\min}$, R_n , and Γ_{opt} are frequently referred to as the noise parameters, and it is their determination which is called noise characterization. When Γ is plotted on a Smith chart for a set of constant noise factors, F , the result is noise circles. Noise circles are a convenient format to display the complex relation between source impedance and noise figure.

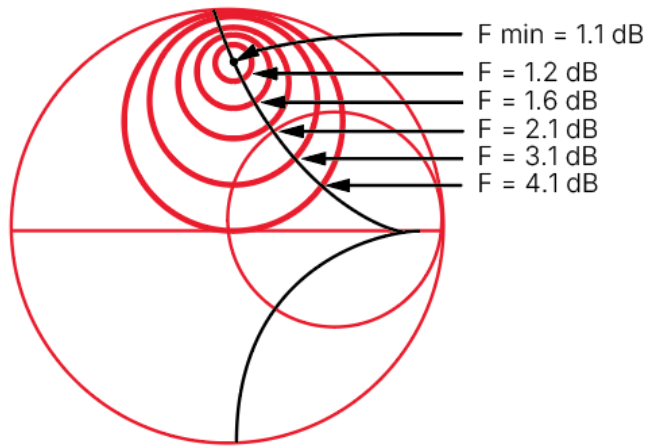


Figure 2-2. Noise circles

The available gain, G_a , provided by a device when it is driven by a specified source impedance, can be calculated from the S-parameters of the device^[35, 40] and the source reflection coefficient, Γ using equation (2-5). S-parameters are commonly measured with a network analyzer.

$$G_a = \frac{(1 - |\Gamma_s|^2) |S_{21}|^2}{|1 - S_{11}\Gamma_s|^2 \left(1 - |S_{22} + \frac{S_{12}S_{21}\Gamma_s}{1 - S_{11}\Gamma_s}|^2 \right)} \quad (2-5)$$

When the source reflection coefficient, Γ_s , is plotted on a Smith chart corresponding to a set of fixed gains, gain circles result. Gain circles are a convenient format to display the relation between source impedance and gain.

The Effect of Bandwidth

Although the system bandwidth is an important factor in many systems and is involved in the actual signal-to-noise calculations for demodulated signals, noise figure is independent of device bandwidth. A general assumption made when performing noise measurements is that the device to be tested has an amplitude-versus-frequency characteristic that is constant over the measurement bandwidth. This means that noise measurement bandwidth should be less than the device bandwidth. When this is not the case, an error will be introduced.^[34] Keysight NF7-class noise figure analyzers provide variable bandwidths to facilitate measurement of narrowband devices.

Most often the bandwidth-defining element in a system, such as a receiver, will be the IF or the detector. It will usually have a bandwidth much narrower than the RF circuits. In this case noise figure is a valid parameter to describe the noise performance of the RF circuitry. In the unusual case where the RF circuits have a bandwidth narrower than the IF or detector, noise figure may still be used as a figure of merit for comparisons, but a complete analysis of the system signal-to-noise ratio will require the input bandwidth as a parameter.

The Measurement of Noise Figure

Noise Power Linearity

The basis of most noise figure measurements depends on a fundamental characteristic of linear two-port devices, noise linearity. The noise power out of a device is linearly dependent on the input noise power or temperature as shown in Figure 3-1. If the slope of this characteristic and a reference point is known, the output power corresponding to a noiseless input power, N_a can be found. From N_a the noise figure or effective input noise temperature can be calculated as described in Chapter 1. Because of the need for linearity, any automatic gain control (AGC) circuitry must be deactivated for noise figure measurements.

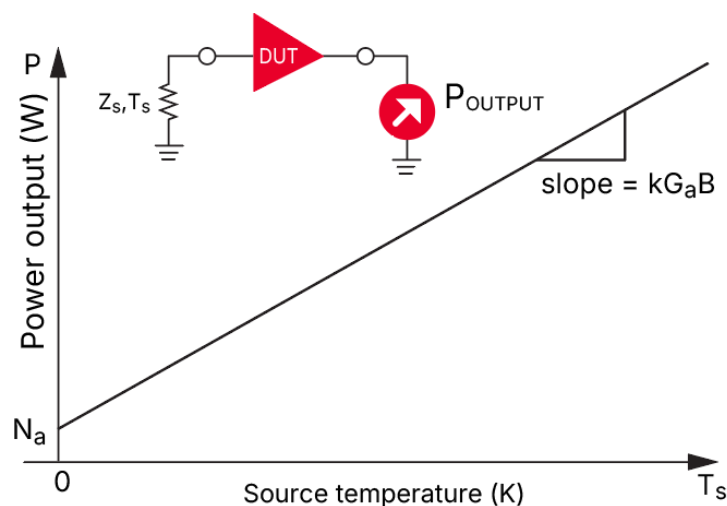


Figure 3-1. The straight-line power output versus source temperature characteristic of linear, two-port devices. For a source impedance with a temperature of absolute zero, the power output consists solely of added noise N_a from the DUT. For other source temperatures the power output is increased by thermal noise from the source amplified by the gain characteristic of the DUT.

Noise Sources

One way of determining the noise slope is to apply two different levels of input noise and measure the output power change. A noise source is a device that will provide these two known levels of noise. The most popular noise source consists of a special low-capacitance diode that generates noise when reverse biased into avalanche breakdown with a constant current^[5]. Precision Keysight noise sources include output attenuation to provide a low SWR and minimize mismatch errors in measurements. If there is a difference between the on and off state impedance an error can be introduced into the noise figure measurement^[23]. Noise sources use output attenuation to improve impedance match and minimize mismatch errors during measurements. Some noise sources provide higher attenuation to further reduce these effects for mismatch-sensitive devices.

When the diode is biased, the output noise will be greater than kT_cB due to avalanche noise generation in the diode^[11, 12, 13, 15, 20, 21]; when unbiased, the output will be the thermal noise produced in the attenuator, kT_cB . These levels are sometimes called T_h and T_c corresponding to the terms hot and cold. Noise sources produce noise levels corresponding to a high effective noise temperature when on and approximately 290 K when off. Keysight noise sources are available across a wide frequency range, including millimeter-wave frequencies, with designs that provide excellent impedance match for accurate measurements.



SNS-Series noise source

To make noise figure measurements a noise source must have a calibrated output noise level, represented by excess noise ratio (ENR). Unique ENR calibration information is supplied with the noise source and, in the case of the SNS-Series, is stored internally on non-volatile memory. Other noise sources may provide calibration data in formats that require manual entry or additional processing. ENR_{dB} is the ratio, expressed in dB of the difference between T_h and T_c , divided by 290 K. It should be noted that a 0 dB ENR noise source produces a 290 K temperature change between its on and off states. ENR is not the on noise relative to kTB as is often erroneously believed.

$$ENR_{dB} = 10 \log \left(\frac{T_h - T_c}{T_0} \right) \quad (3-1)$$

T_c in equation (3-1) is assumed to be 290 K when it is calibrated. When the noise source is used at a different physical temperature, compensation must be applied to the measurement. The SNS-Series noise sources contain a temperature sensor which can be read by Keysight's analyzers. The temperature compensation will be covered in the next section of this chapter.

In many noise figure calculations, the linear form of ENR will be used.

$$ENR = 10^{\frac{ENR_{dB}}{10}} \quad (3-2)$$

Noise sources may be calibrated from a transfer standard noise source (calibrated traceable to a top-level National Standards laboratory) or by a primary physical standard such as a hot/ cold load. Most noise sources will be supplied with an ENR characterized versus frequency.

Hot and cold loads are used in some special applications as a noise source. Ideally the two loads need to be kept at constant temperatures for good measurement precision. One method immerses one load into liquid nitrogen at a temperature of 77 K, the other may be kept at room temperature or in a temperature-controlled oven. The relatively small temperature difference compared to noise diode sources and potential SWR changes resulting from switching to different temperature loads usually limits this method to calibration labs and millimeter-wave users.

Gas discharge tubes embedded in waveguide structures produce noise due to the kinetic energy of the plasma and were historically used as millimeter-wave noise sources. In modern systems, solid-state noise diodes and related technologies are widely used across much of the RF and microwave frequency range. The noise diode is simpler to use and generally is a more stable source of noise. Although the noise diode is generally a coaxial device, integral, precision waveguide adapters may be used to provide a waveguide output.



Millimeter-wave noise sources

The Y-Factor Method

The Y-Factor method is the basis of most noise figure measurements whether they are manual or automatically performed internally in a noise figure analyzer. Using a noise source, this method allows the determination of the internal noise in the DUT and therefore the noise figure or effective input noise temperature.

With a noise source connected to the DUT, the output power can be measured corresponding to the noise source on and the noise source off (N_2 and N_1). The ratio of these two powers is called the Y-factor. The power detector used to make this measurement may be a power meter, spectrum analyzer, or a special internal power detector in the case of noise figure meters and analyzers. The relative level accuracy is important. One of the advantages of modern noise figure analyzers is that the internal power detector is very linear and can very precisely measure level changes. The absolute power level accuracy of the measuring device is not important since a ratio is to be measured.

$$Y = \frac{N_2}{N_1} \quad (3-3)$$

Sometimes this ratio is measured in dB units, in this case:

$$Y = 10^{\frac{Y_{dB}}{10}} \quad (3-4)$$

The Y-factor and the ENR can be used to find the noise slope of the DUT that is depicted in Figure 3-1. Since the calibrated ENR of the noise source represents a reference level for input noise, an equation for the DUT internal noise, N_a can be derived. In modern noise figure measurement systems, this is automatically determined by modulating the noise source between the on and off states and applying internal calculations.

$$N_a = kT_0BG_1 \left(\frac{\text{ENR}}{Y-1} - 1 \right) \quad (3-5)$$

From this we can derive a very simple expression for the noise factor. The noise factor that results is the total system noise factor, F_{sys} . System noise factor includes the noise contribution of all the individual parts of the system. In this case the noise generated in the measuring instrument has been included as a second stage contribution. If the DUT gain is large ($G_1 \gg F_2$), the noise contribution from this second stage will be small. The second stage contribution can be removed from the calculation of noise figure if the noise figure of the second stage and the gain of the DUT is known. This will be covered in the section on corrected noise figure and gain. Note that the device gain is not needed to find F_{sys} .

$$F_{\text{sys}} = \frac{\text{ENR}}{Y-1} \quad (3-6)$$

When the noise figure is much higher than the ENR, the device noise tends to mask the noise source output. In this case the Y-factor will be very close to 1. Accurate measurement of small ratios can be difficult. Generally, the Y-factor method is not used when the noise figure is more than 10 dB above the ENR of the noise source, depending on the measurement instrument.

This equation can be modified to correct for the condition when the noise source cold temperature, T_c , is not at the 290 K reference temperature, T_0 .

$$F_{\text{sys}} = \frac{\text{ENR} - Y \left(\frac{T_c}{T_0} - 1 \right)}{Y-1} \quad (3-7)$$

This often-used equation assumes that T_h is unaffected by changes in T_c as is the case with hot and cold loads. With solid-state noise sources, T_h will likely be affected by changes in T_c . Since the physical noise source is at a temperature of T_c , the internal attenuator noise due to T_c is added both when the noise source is on and off. In this case it is better to assume that the noise change between the on and off state remains constant ($T_h - T_c$). This distinction is most important for low ENR noise sources when T_h is less than 10 T_c .

An alternate equation can be used to correct for this case.

$$F_{\text{sys}} = \frac{\text{ENR} \left(\frac{T_c}{T_0} \right)}{Y-1} \quad (3-8)$$

The Signal Generator Twice-Power Method

Before calibrated noise sources were widely available, this method was commonly used for noise figure measurements. It is still particularly useful for high noise figure devices where the Y-factors can be very small and difficult to accurately measure. First, the output power is measured with the device input terminated with a load at a temperature of approximately 290 K. Then a signal generator is connected, providing a signal within the measurement bandwidth. The generator's output power is adjusted to produce a 3 dB increase in the output power. If the generator power level and measurement bandwidth are known, we can calculate the noise factor. It is not necessary to know the DUT gain.

There are some factors that limit the accuracy of this method.

$$F_{\text{sys}} = \frac{P_{\text{gen}}}{kT_0 B} \quad (3-9)$$

The noise bandwidth of the power-measuring device must be known, perhaps requiring a network analyzer. Noise bandwidth, B , is a calculated equivalent bandwidth, having a rectangular, "flat-top" spectral shape with the same gain bandwidth product as the actual filter shape. The output power must be measured on a device that measures true power since we have a mix of noise and a CW signal present. Thermal-based power meters measure true power accurately; however, when used in low-level noise measurements, they may require additional amplification and bandwidth-defining filtering. Spectrum analyzers have good sensitivity and a well-defined bandwidth, but the detector may respond differently to CW signals and noise. Absolute level accuracy is not needed in the power detector since a ratio is being measured.

The Direct Noise Measurement Method

This method is also useful for high noise figure devices. The output power of the device is measured with an input termination at a temperature of approximately 290 K. If the gain of the device and noise bandwidth of the measurement system is known, the noise factor can be determined.

$$F_{\text{sys}} = \frac{N_o}{kT_0 B G} \quad (3-10)$$

Again, with this method the noise bandwidth, B , must be known and the power-measuring device may need to be very sensitive. Unlike the twice-power method, the DUT gain must be known, and the power detector must have absolute level accuracy.

Corrected Noise Figure and Gain

The previous measurements are used to measure the total system noise factor, F_{sys} , including the measurement system. Generally, it is the DUT noise figure that is desired. From the cascade noise-figure equation it can be seen that if the DUT gain is large, the measurement system will have little effect on the measurement. The noise figure of high gain DUTs can be directly measured with the previously discussed methods. When a low gain DUT is to be measured or the highest accuracy is needed, a correction can be applied if we know the gain of the DUT and the noise figure of the system. Using equation (2-2) and re-writing to solve for F_1 gives the equation for the actual DUT noise factor.

$$F_1 = F_{sys} - \frac{F_2 - 1}{G_1} \quad (3-11)$$

Both the gain of the DUT and the measurement system noise factor, F_2 , can be determined with an additional noise source measurement. This step is called a system calibration. In noise figure measurement systems, this calibration is typically performed before connecting the DUT so that subsequent measurements can use the corrections and display corrected noise figure. The necessary calculations to find the gain and the corrected noise figure are automatically performed internally. When manual measurements are made with alternative instruments, a calibrated noise figure measurement can be performed as follows:

1. Connect the noise source directly to the measurement system and measure the noise power levels corresponding to the noise source "on" and "off". These levels; N_2 and N_1 respectively, can then be used to calculate the measurement system noise factor F_2 using the Y-factor method.
2. The DUT is inserted into the system. The noise levels N_2 and N_1 are measured when the noise source is turned on and off. The DUT gain can be calculated with the noise level values.

$$G_1 = \frac{N_2' - N_1'}{N_2 - N_1} \quad (3-12)$$

The gain is usually displayed in dB terms:

$$G_{db} = 10\log G$$

3. The overall system noise factor, F_{sys} , can be calculated by applying the Y-factor method to the values N_2' and N_1' .
4. The DUT noise factor, F_1 , can be calculated with equation (3-11). The DUT noise figure is $10\log F_1$.

Jitter

Noise can be thought of as a series of random events, electrical impulses in this case. The goal of any noise measurement is to find the mean noise level at the output of the device. These levels can be used, with appropriate corrections, to calculate the actual noise figure of the device. In theory, the time required to find the true mean noise level would be infinite. In practice, averaging is performed over some finite time period. The difference between the measured average and the true mean will fluctuate and give rise to a repeatability error.

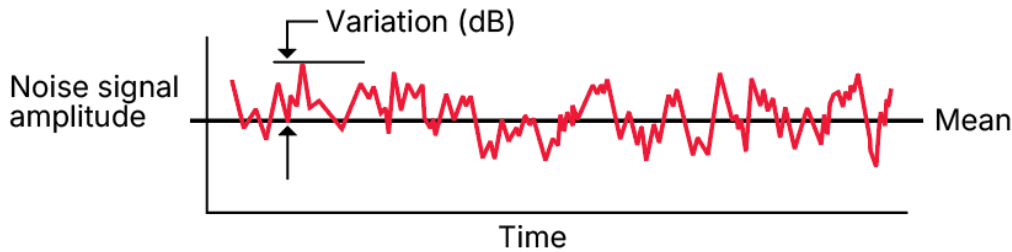


Figure 3-2. Noise jitter

For small variations, the deviation is proportional to $1/\sqrt{(t)}$ so that longer averaging times will produce better averages. Because the average includes more events it is closer to the true mean. The variation is also proportional to $1/\sqrt{(B)}$. Larger measurement bandwidths will produce a better average because there are more noise events per unit of time in a large bandwidth; therefore, more events are included in the average. Usually, noise figure should be measured with a bandwidth as wide as possible but narrower than the DUT.

Frequency Converters

Frequency converters such as receivers and mixers usually are designed to convert an RF frequency band to an IF frequency band. While the noise figure relationships discussed in this application note apply to converters as well as non-converters, there are some additional characteristics of these devices that can affect noise figure measurements. In addition to DUTs that are frequency converters, sometimes the noise measurement system uses mixing to extend the measurement frequency range.

Loss

Amplifiers usually have a gain associated with them, while passive mixers have loss. All the equations for noise figure still apply; however, the linear gain values used will be less than one. One implication of this can be seen by applying the cascade noise figure equation; the second stage noise contribution can be major (See Equation 2-2). Another is that passive mixers, if measured using the Y-factor technique, can have small Y-factors owing to their high noise figures. This may increase measurement uncertainty. High ENR noise sources can be used to provide a larger Y-factor.

LO Noise

Receivers and mixers have local oscillator (LO) signals that may have noise present. This noise can be converted in the mixer to the IF frequency band and become an additional contribution to the system's noise figure. The magnitude of this effect varies widely depending on the specific mixer type and how much noise is in the LO. In some systems, LO-related noise can be reduced by filtering the LO signal, for example using a band-pass filter at the LO port of the mixer; however, the effectiveness of this approach depends on the LO architecture and bandwidth, particularly in modern agile or wideband systems. There may also be higher order noise conversions that could contribute if the LO noise level is very high. A lowpass filter can be used to prevent noise conversions at harmonics of the LO frequency.

LO Leakage

A residual LO signal may be present at the output (IF) of a mixer or converter. The presence of this signal is generally unrelated to the noise performance of the DUT and may be acceptable when used for the intended application. When a noise figure measurement is made, this LO signal may overload the noise measurement instrument or create other spurious mixing products. This is most likely to be an issue when the measuring system has a broadband amplifier or other unfiltered circuit at its input. Often a filter can be added to the instrument input to filter out the LO signal while passing the IF.

Unwanted Responses

Sometimes the desired RF frequency band is not the only band that converts to the IF frequency band. Unwanted frequency band conversions may occur if unwanted frequencies are present at the RF port in addition to the desired RF signal. Some of these are: the image response ($f_{LO} + f_{IF}$ or $f_{LO} - f_{IF}$ depending on the converter), harmonic responses ($2f_{LO} \pm f_{IF}$, $3f_{LO} \pm f_{IF}$, etc.), spurious responses, and IF feed-through response. Often, particularly in receivers, these responses are negligible due to internal filtering. With many other devices, especially mixers, one or more of these responses may be present and may convert additional noise to the IF frequency band.

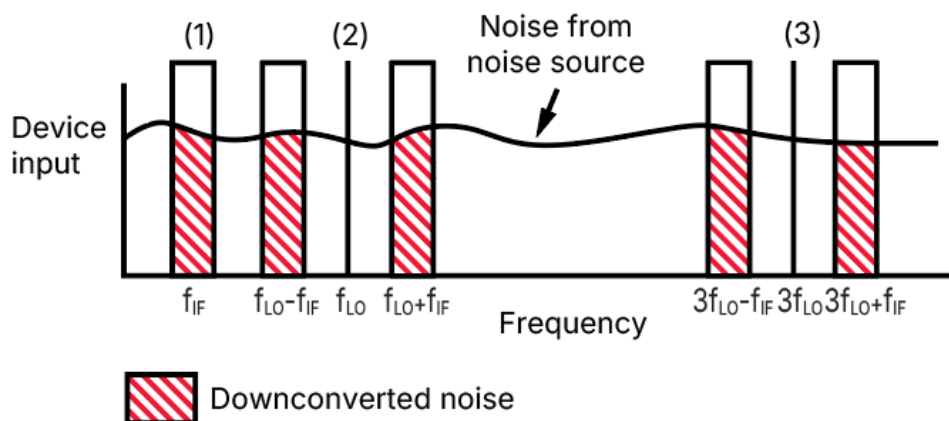


Figure 3-3. Possible noise conversion mechanisms with mixers and converters. (1) IF feedthrough response, (2) double sideband response, (3) harmonic response.

Mixers having two main responses ($f_{LO} + f_{IF}$ and $f_{LO} - f_{IF}$) are often termed double sideband (DSB) mixers. $f_{LO} + f_{IF}$ is called the upper sideband (USB). $f_{LO} - f_{IF}$ is called the lower sideband (LSB). They convert noise in both frequency bands to the IF frequency band. When such a mixer is part of the noise measurement system, the second response will create an error in noise figure measurements unless a correction, usually +3 dB, is applied. Ideally filtering is used at the RF port to eliminate the second response so that single sideband (SSB) measurements can be made.

When a DSB mixer is the DUT, we have a choice when measuring the noise figure. Usually, the user wants to measure the equivalent SSB noise figure. In passive mixers that do not have LO noise, the equivalent SSB noise figure is often close in value to the conversion loss measured with a CW signal. There are two ways to make this measurement; an input filter can be used, or the +3 dB correction can be applied. There are accuracy implications with these methods that must be considered if precision measurements are to be made; an input filter will add loss that should be corrected for, the +3 dB correction factor assumes equal USB and LSB responses.

Converters used in noise receivers, such as radiometers and radiometric sensors, are often designed to make use of both main responses, in which case it is desirable to know the DSB noise figure. In this case, no correction or input filter is used; the resulting noise figure measured will be in DSB terms.

Noise Figure Measuring Instruments

Noise Figure Analyzers

Noise figure analyzers are dedicated instruments for noise figure measurements, complementing signal analyzers with integrated noise figure measurement applications. Modern noise figure measurement systems include a receiver with accurate power detection, noise source control, and integrated processing to determine noise figure. Calibration data, such as ENR, is typically handled automatically, and results are displayed as a function of frequency. Internally, these systems commonly compute noise figure using the Y-factor method.

A noise figure analyzer allows the display of swept frequency noise figure and gain and associated features such as markers and limit lines.



NF7-class noise figure analyzer

Signal/Spectrum Analyzers

Signal and spectrum analyzers are often used to measure noise figure because they are already present in many RF and microwave test environments. With integrated measurement applications and automated control, they can be used to measure noise figure using the methods outlined in this application note. They are particularly useful for measuring high noise figure devices using the signal generator or direct power measurement method. With the available noise figure measurement application, they can achieve uncertainties negligibly degraded from the capabilities of the noise source used. The variable resolution bandwidths allow measurement of narrow-band devices.

One of the advantages of a signal/spectrum analyzer is multi-functionality. Typically, measurement applications may be added to make measurements specific to a particular communications standard.



Essential grade signal analyzer

Network Analyzers

Like spectrum analyzers, network analyzers are common multi-use instruments. Modern vector network analyzers (VNAs) support integrated noise figure measurements alongside traditional network measurements, using dedicated applications and calibration techniques. An advantage is that they can offer other measurements commonly associated with devices: such as gain and match. Because network measurements are usually made with the same internal receiver architecture, there can be some performance limitations when used in noise figure applications. Often the receiver is of the double sideband (DSB) type, where noise figure is actually measured at two frequencies and an internal correction is applied. When a wide measurement bandwidth is used this may result in error if the device noise figure or gain is not constant over this frequency range. When narrow measurement bandwidth is used to measure narrow-band devices, the unused frequency spectrum between the upper and lower sideband does not contribute to the measurement, and a longer measurement time is needed to reduce jitter (see Jitter in this chapter).

Network analyzers have the ability to measure the S-parameters of the device. It has been considered that S-parameter data can reduce noise figure measurement uncertainty by offering mismatch correction. Ideally this mismatch correction would provide a more accurate gain measurement of the device so that the second stage noise contribution can be subtracted with more precision. Unfortunately, the mismatch also affects the noise generation in the second stage which cannot be corrected for without knowing the noise parameters of the device. The same situation occurs at the input of the device when a mismatch is present between the noise source and DUT input. (see Noise parameters in Chapter 2)^[4]. Network analyzers do not, by themselves, provide measurement of the noise parameters. Measurement of noise parameters generally requires a tuner and software in addition to the network analyzer. The resulting measurement system can be complex and expensive. Error correction in a network analyzer is primarily of benefit for gain measurements and calculation of available gain.

Noise Parameter Test Sets

Noise parameter test systems are typically used in conjunction with a vector network analyzer and noise figure measurement instrumentation, along with coordinated control and processing, to perform a series of measurements that enable determination of device noise parameters^[29] (see Noise parameters in Chapter 2). Noise parameters can then be used to calculate the minimum device noise figure, the optimum source impedance, and the effect of source impedance on noise figure. The test set has an adjustable tuner to present various source impedances to the DUT. Internal networks provide bias to semiconductor devices that may be tested. A noise source is coupled to the test set to allow noise figure measurements at different source impedances. The corresponding source impedances are measured with the network analyzer. From this data, the complete noise parameters of the device can be calculated. Generally, the complete device S-parameters are also measured so that gain parameters can also be determined. Because of the number of measurements involved, measurement of the full noise parameters of a device is much slower than making a conventional noise figure measurement but yields useful design parameters. Noise parameters are often supplied on low-noise transistor data sheets. Noise parameters are generally not measured on components and assemblies that are intended to be used in well matched 50 Ω (or 75 Ω) stems because the source impedance is defined in the application.

Power Meters and True-RMS Voltmeters

Basic instruments such as power meters and true-RMS voltmeters can be used to measure noise figure using the methods described in this note; however, this approach typically requires manual or external processing and is primarily of instructional or specialized use. Being broadband devices, they need a filter to limit their bandwidth to be narrower than the DUT. Such a filter will usually be fixed in frequency and allow measurements only at this frequency. Power meters are most often used to measure receiver noise figures where the receiver has a fixed IF frequency and much gain. The sensitivity of power meters and voltmeters is usually poor, but the receiver may provide enough gain to make measurements. If additional gain is added ahead of a power meter to increase sensitivity, care should be taken to avoid temperature drift and oscillations.



Peak power meter

Glossary

Symbols and Abbreviations

B	Noise bandwidth
BER	Bit error ratio
$ b_s ^2$	Power delivered by a generator to a non-reflecting load
C/N	Carrier to noise ratio
DBS	Direct broadcast by satellite
DSB	Double sideband
DUT	Device under test
ENR _{dB}	Excess noise ratio
F	Noise factor
F ₁	First stage noise factor
FM	Frequency modulation
F _{min}	Minimum noise factor
F _{sys}	System noise factor
1/f	Flicker noise
G _p	Power gain
G _{ass}	Associated gain
G _a	Available gain
G _i	Insertion gain
G _t	Transducer gain
G/T	Gain-to-temperature ratio
IEEE	Institute of electrical and electronics engineers
IF	Intermediate frequency
IRE	IRE Institute of Radio Engineers (merged into IEEE)
K	Kelvins (unit of temperature)
k	Boltzmann's constant
LNA	Low noise amplifier
LSB	Lower sideband
M	Noise measure
M _u	Mismatch uncertainty
N _a	Noise added
NF	Noise figure
N _{off}	=N ₁ (see Y factor)
N _{on}	=N ₂ (see Y factor)
N ₁	Nout for T _c (see Y factor)
N ₂	Nout for T _h (see Y factor)
N _i	Input noise power
N _o	Output Noise power
RF	Radio frequency
RMS	Root mean square
R _n	Equivalent noise resistance
r _n	Equivalent Noise resistance, normalized
RSS	Root Sum-of-the-squares
S/N	Signal to noise ratio
SSB	Single sideband
$ S_{21} ^2$	Forward transmission coefficient
S _i	Input signal power
S _o	Output signal power
T _a	Noise temperature
T _c	Cold temperature (see T _c)
T _e	Effective input noise temperature
T _h	Hot temperature (see T _h)
T _{ne}	Effective noise temperature
T _{off}	Off temperature (see T _{off})

Glossary Terms

Associated gain (G_{ass}). The available gain of a device when the source reflection coefficient is the optimum reflection coefficient Γ_{opt} corresponding with F_{min} .

Available gain (G_a). [2, 35, 40] The ratio, at a specific frequency, of power available from the output of the network P_{ao} to the power available from the source P_{as} .

$$G_a = \frac{P_{ao}}{P_{as}} \quad (1)$$

For a source with output $|b_s|^2$ and reflection coefficient Γ

$$P_{as} = \frac{|b_s|^2}{1 - |\Gamma_s|^2} \quad (2)$$

$$P_{ao} = \frac{|b_s|^2 |S_{21}|^2 (1 - |\Gamma_2|^2)}{|(1 - \Gamma_s S_{11})(1 - \Gamma_2^* S_{22}) - \Gamma_s \Gamma_2^* S_{12} S_{21}|^2} \quad (3)$$

where

$$T_2 = S_{22} + \frac{S_{12} S_{21} \Gamma_s}{1 - S_{11} \Gamma_s} \quad (4)$$

An alternative expression for the available output power is

$$P_{ao} = \frac{|b_s|^2 |S_{21}|^2}{|1 - \Gamma_s S_{11}|^2 (1 - |\Gamma_2|^2)} \quad (5)$$

These lead to two expressions for G_a

$$G_a = |S_{21}|^2 \frac{(1 - |\Gamma_s|^2) (1 - |\Gamma_2|^2)}{|(1 - \Gamma_s S_{11})(1 - \Gamma_2^* S_{22}) - \Gamma_s \Gamma_2^* S_{12} S_{21}|^2} \quad (6)$$

$$G_a = |S_{21}|^2 \frac{1 - |\Gamma_s|^2}{|1 - \Gamma_s S_{11}|^2 (1 - |\Gamma_2|^2)} \quad (7)$$

NOTE: G_a is a function of the network parameters and of the source reflection coefficient Γ_L . G_a is independent of the load reflection coefficient Γ_L . G_a is often expressed in dB

$$G_a(\text{dB}) = 10 \log \frac{P_{ao}}{P_{as}} \quad (8)$$

Bandwidth (B). See noise bandwidth.

Boltzmann's constant (k). 1.38×10^{-23} joules/kelvin.

Cascade effect.^[8] The relationship, when several networks are connected in cascade, of the noise characteristics (F or T_e and G_a) of each individual network to the noise characteristics of the overall or combined network.

If $F_1, F_2, \dots, F_n$ (numerical ratios, not dB) are the individual noise figures and $G_{a1}, G_{a2}, \dots, G_{an}$ (numerical ratios) are the individual available gains, the combined noise figure is

$$F = F_1 + \frac{F_2 - 1}{G_{a1}} + \frac{F_3 - 1}{G_{a1}G_{a2}} + \dots + \frac{F_n - 1}{G_{a1}G_{a2}\dots G_{a(n-1)}} \quad (1)$$

the combined available gain is

$$G_a = G_{a1}G_{a2}\dots G_{an} \quad (2)$$

In terms of individual effective input noise temperatures $T_{e1}, T_{e2}, \dots, T_{en}$ the overall effective input noise temperature is

$$T_e = T_{e1} + \frac{T_{e2}}{G_{a1}} + \frac{T_{e2}}{G_{a1}G_{a2}} + \dots + \frac{T_{en}}{G_{a1}G_{a2}\dots G_{a(n-1)}} \quad (3)$$

NOTE: Each F_i , T_{ei} , and G_{ai} above refers to the value for the source impedance that corresponds to the output impedance of the previous stage.

Diode noise source. ^[11, 12, 13, 15, 20, 21] A noise source that depends on the noise generated in a solid state diode that is reverse biased into the avalanche region. Excess noise ratios of well-matched devices are usually about 15 dB ($T_{ne} = 10000$ K). Higher excess noise ratios are possible by sacrificing impedance match and flat frequency response.

Double sideband (DSB). See Single-sideband (SSB).

Effective input noise temperature (T_e).^[17] The noise temperature assigned to the impedance at the input port of a DUT which would, when connected to a noise-free equivalent of the DUT, yield the same output power as the actual DUT when it is connected to a noise-free input port impedance. The same temperature applies simultaneously for the entire set of frequencies that contribute to the output frequency. If there are several input ports, each having a specified impedance, the same temperature applies simultaneously to all the ports. All ports except the output are to be considered input ports for purposes of defining T_e . For a two-port transducer with a single input and a single output frequency, T_e is related to the noise figure F by

$$T_e = 290 (F - 1) \quad (1)$$

Effective noise temperature (T_{ne}).^[1] (This is a property of a one-port, for example, a noise source.) The temperature that yields the power emerging from the output port of the noise source when it is connected to a nonreflecting, nonemitting load. The relationship between the noise temperature T_a and effective noise temperature T_{ne} is

$$T_{ne} = T_a(1 - |\Gamma|^2) \quad (1)$$

where Γ is the reflection coefficient of the noise source. The proportionality factor for the emerging power is kB so that

$$T_{ne} = \frac{P_e}{(kB)} \quad (2)$$

where P_e is the emerging power, k is Boltzmann's constant, and B is the bandwidth of the power measurement. The power spectral density across the measurement bandwidth is assumed to be constant.

Equivalent noise resistance (r_n or R_n). See noise figure circles.

Excess noise ratio (ENR).^[1] A noise generator property calculated from the hot and cold noise temperatures (T_h and T_c) using the equation

$$ENR_{dB} = 10 \log \frac{T_h - T_c}{T_0} \quad (1)$$

where T_0 is the standard temperature of 290 K. Noise temperatures T_h and T_c should be the "effective" noise temperatures. (See Effective Noise Temperature)^[25]. The ENR calibration of diode noise sources assumes $T_c = T_0$.

A few examples of the relationship between ENR and T_h may be worthwhile. An ENR of 0 dB corresponds to $T_h = 580$ K. T_h of 100°C (373 K) corresponds to an ENR of -5.43 dB. T_h of 290 K corresponds to an ENR of $-\infty$ dB.

Flicker noise and 1/f noise. ^[33, 39] Any noise whose power spectral density varies inversely with frequency. Especially important at audio frequencies or with GASFET's below about 100 MHz.

Forward transmission coefficient (S_{21}).² The ratio, at a specific frequency, of the power delivered by the output of a network, to the power delivered to the input of the network when the network is terminated by a nonreflecting load and excited by a nonreflecting generator.

The magnitude of this parameter is often given in dB.

$$|S_{21}|^2 \text{ (dB)} = 10 \log |S_{21}|^2 \quad (1)$$

Gain to temperature ratio (G/T). ^[32, 41] A figure of merit for a satellite or radio astronomy receiver system, including the antenna, that portrays the operation of the total system. The numerator is the antenna gain, the denominator is the operating noise temperature of the receiver. The ratio is usually expressed in dB, for example, $10\log(G/T)$. G/T is often measured by comparing the receiver response when the antenna input is a hot celestial noise source to the response when the input is the background radiation of space (3K).

Gas discharge noise source. ^[25, 26] A noise source that depends on the temperature of an ionized noble gas. This type of noise source usually requires several thousand volts to begin the discharge but only about a hundred volts to sustain the discharge. Components of the high turn-on voltage sometimes feed through the output to damage certain small, frail, low-noise, solid-state devices. The gas discharge noise source has been replaced by the avalanche diode noise source in most applications. Gas discharge tubes are still used at millimeter wavelengths. Excess noise ratios (ENR) for argon tubes is about 15.5 dB (10000 K).

Gaussian noise. ^[6] Noise whose probability distribution or probability density function is gaussian, that is, it has the standard form

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} \quad (1)$$

where σ is the standard deviation. Noise that is steady or stationary in character and originates from the sum of a large number of small events, tends to be gaussian by the central limit theorem of probability theory. Thermal noise and shot noise are gaussian.

Hot/cold noise source. In one sense most noise figure measurements depend on noise power measurements at two source temperatures—one hot and one cold. The expression Hot/Cold, however, frequently refers to measurements made with a cold termination at liquid nitrogen temperatures (77 K) or even liquid helium (4 K), and a hot termination at 373 K (100°C). Such terminations are sometimes used as primary standards and for highly accurate calibration laboratory measurements.

Insertion gain (G_i). The gain that is measured by inserting the DUT between a generator and load. The numerator of the ratio is the power delivered to the load while the DUT is inserted, P_d . The denominator, or reference power P_r , is the power delivered to the load while the source is directly connected. Measuring the denominator might be called the calibration step.

$$G_i = \frac{P_d}{P_r} \quad (1)$$

The load power while the source and load are directly connected is

$$P_r = |b_r|^2 \frac{1 - |\Gamma_1|^2}{|1 - \Gamma_1 \Gamma_s|^2} \quad (2)$$

where the subscript r denotes the source characteristics while establishing the reference power, i.e., during the calibration step. The load power while the DUT is inserted is

$$P_d = |b_d|^2 |S_{21}|^2 \frac{1 - |\Gamma_1|^2}{|(1 - \Gamma_{sd} S_{11})(1 - \Gamma_1 S_{22}) - \Gamma_1 S_{12} S_{21}|^2} \quad (3)$$

or

$$P_d = |b_d|^2 |S_{21}|^2 \frac{1 - |\Gamma_1|^2}{|1 - \Gamma_{sd} S_{11}|^2 |1 - \Gamma_1 \Gamma_2|^2} \quad (4)$$

$$T_2 = S_{22} + \frac{S_{12} S_{21} \Gamma_{sd}}{1 - \Gamma_{sd} S_{11}} \quad (5)$$

In equations (3, 4, and 5) the subscript d denotes the source characteristics while the DUT is inserted. The S-parameters refer to the DUT. The source characteristics while calibrating and while the DUT is inserted are sometimes different. Consider that the DUT, for example, is a microwave receiver with a waveguide input and an IF output at 70 MHz. During the calibration step, the source has a coaxial output at 70 MHz, but while the DUT is inserted the source has a waveguide output at the microwave frequency. Using the above equations, insertion gain is

$$G_i = |S_{21}|^2 \frac{|b_d|^2}{|b_r|^2} \frac{|1 - \Gamma_1 \Gamma_{sr}|^2}{|(1 - \Gamma_{sd} S_{11})(1 - \Gamma_1 S_{22}) - \Gamma_1 \Gamma_{sd} S_{12} S_{21}|^2} \quad (6)$$

$$G_i = |S_{21}|^2 \frac{|b_d|^2}{|b_r|^2} \frac{|1 - \Gamma_1 \Gamma_{sr}|^2}{|1 - \Gamma_{sd} S_{11}|^2 |1 - \Gamma_1 \Gamma_2|^2} \quad (7)$$

In those situations where the same source at the same frequency is used during the calibration step and DUT insertion, $|b_d|^2 = |b_r|^2$ and $G_{sr} = G_{sd}$. This is usually the case when measuring amplifiers.

Instrument uncertainty. The uncertainty caused by errors within the circuits of electronic instruments. For noise figure analyzers/meters this includes errors due to the detector, A/D converter, math round-off effects, any mixer non-linearities, saturation effects, and gain instability during measurement. This uncertainty is often mistakenly taken as the overall measurement accuracy because it can be easily found on specification sheets. With modern techniques, however, it is seldom the most significant cause of uncertainty.

Johnson noise. ^[19] The same as thermal noise.

Minimum noise factor ($F_{\min}$). See Noise Figure Circles.

Mismatch uncertainty (M_u). Mismatch uncertainty is caused by re-reflections between one device (the source) and the device that follows it (the load). The re-reflections cause the power emerging from the source (incident to the load) to change from its value with a reflectionless load.

An expression for the power incident upon the load, which includes the effects of re-reflections, is

$$P_i = \frac{|b_s|^2}{|1 - \Gamma_s \Gamma_l|^2} \quad (1)$$

where $|b_s|^2$ is the power the source delivers to a non-reflecting load, Γ_s is the source reflection coefficient, and Γ_l is the load reflection coefficient. If accurate evaluation of the power incident is needed when $|b_s|^2$ is given or vice versa, then the phase and magnitude of Γ_s and Γ_l is needed—probably requiring a vector network analyzer.

When the phase of the reflection coefficients is not known, the extremes of $|1 - \Gamma_s \Gamma_l|^2$ can be calculated from the magnitudes of Γ_s and Γ_l , for example, P_s and P_l . The extremes of $|1 - \Gamma_s \Gamma_l|^2$ in dB can be found from the nomograph (Figure 4-1).

$$M_u = 20 \log(1 \pm P_s P_l)$$

The effect of mismatch on noise figure measurements is extremely complicated to analyze. Consider, for example, a noise source whose impedance is not quite 50 Ω .

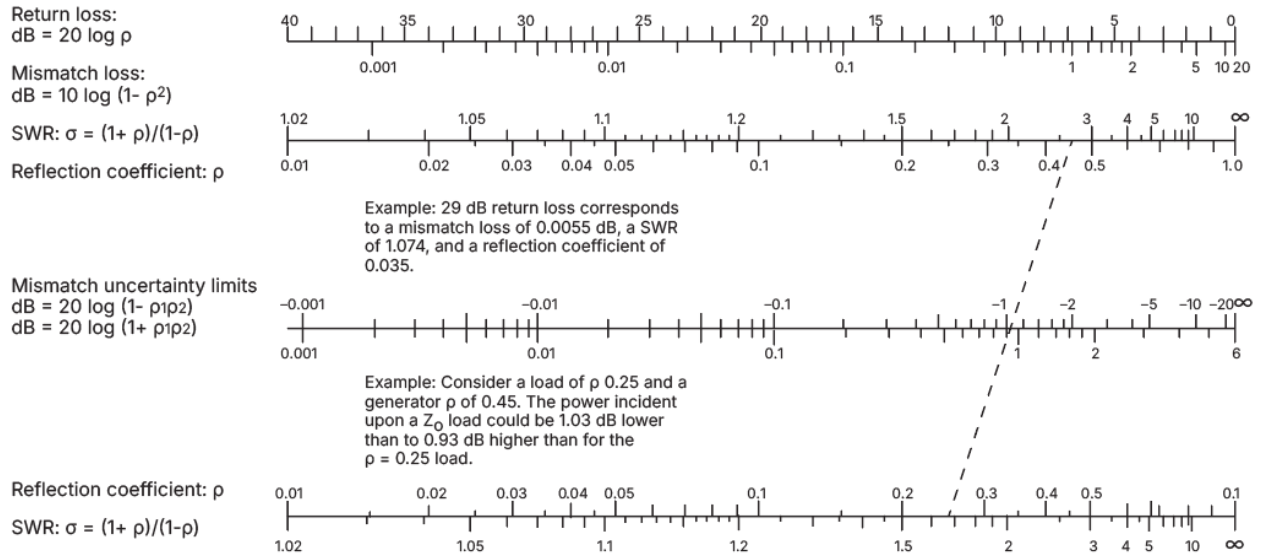


Figure 4-1. This nomograph gives the extreme effects of re-reflections when only the reflection coefficient magnitudes are known. Mismatch uncertainty limits of this nomograph apply to noise figure measurement accuracy for devices that include an isolator at the input.

The source takes part in re-reflections of its own generated noise, but it also reflects noise originating in the DUT and emerging from the DUT input (noise added by a DUT, after all, is a function of the source impedance). The changed source impedance also causes the DUT's available gain to change (remember that available gain is also a function of source impedance). The situation can be complicated further because the source impedance can change between the hot state and the cold state. ^[23] Many attempts have been made to establish a simple rule-of-thumb for evaluating the effect of mismatch—all with limited success. One very important case was analyzed by Strid ^[36] to have a particularly simple result. Strid considered the DUT to include an isolator at the input with sufficient isolation to prevent interaction of succeeding devices with the noise source. The effect of noise emerging from the isolator input and re-reflections between the isolator and noise source are included in the final result. The result is that the error in noise figure is

where F_{act} is the noise figure for a reflectionless noise source,

$$\Delta F(\text{dB}) = F_{act}(\log) - F_{ind}(\text{dB}) \quad (2)$$

$$\begin{aligned} \Delta F(\text{dB}) &= F_{act}(\log) - F_{ind}(\text{dB}) \\ &= 10 \log \frac{1}{|1 - S_{11} \Gamma_{sh}|^2} \end{aligned} \quad (3)$$

F_{ind} is the measured noise figure, S_{11} is the reflection coefficient looking into the DUT, for example, into the isolator input, and G_{sh} is the reflection coefficient looking back into the noise source when in the hot or on condition. Strid also assumed that the isolator and T_{cold} are both 290 K. Note that the result is independent of the DUT noise figure, Y factor, and the noise source reflection coefficient for T_{cold} .

Mismatch uncertainty may also occur while characterizing the noise contribution of the measurement system and also at the output of DUT during gain measurement. Gain measurement mismatch effects can be calculated by evaluating the difference between available gain and insertion gain.

Mismatch uncertainty is often the most significant uncertainty in noise figure measurements. Correction usually requires full noise characterization (see noise figure circles) and measurement of phase and amplitude of the reflection coefficients.

N_1	See Y factor.
N_2	See Y factor.
N_{off}	Same as N_1 . See Y factor.
N_{on}	Same as N_2 . See Y factor.

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Related Literatures and Additional Resources

1. *10 Hints for Making Successful Noise Figure Measurements*, Application Note, literature number [5980-0288E](#)
2. *Noise Figure Measurement Accuracy*, Application Note, literature number [5952-3706E](#)

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